CONCLUSION

The primary objective of this project was to execute the planning, structural analysis, and design single-story (G+0) residential building. This was conducted specifically for the Government of Telangana's 'Indiramma Illu' housing scheme, which aims to provide safe and affordable housing for the poor. By applying core civil engineering principles, this project successfully bridges the gap between stringent economic constraints and the necessity for safe, durable human habitation.

What defines the success of this project is its rigorous engineering methodology: the entire structural framework was calculated and designed completely manually, without reliance on automated structural software. Based on actual site data gathered from a shortlisted beneficiary's plot in Mannapur Village , a highly optimized 590 sq. ft. plinth area was planned. This layout perfectly fulfills the scheme's mandate while maximizing spatial utility for the residents.

The structural analysis was systematically executed using Kani's Method, allowing for the accurate determination of moments across the continuous beams and columns. Subsequently, the Limit State Method, in strict accordance with IS 456:2000, was utilized to design all vital structural components:

- **Roof Slab:** Designed safely as a two-way slab with an overall depth of 100 mm. It was mathematically proven to be safe in both bending and deflection.
- **Beams & Columns:** Meticulously detailed using M20 grade concrete and Fe500 steel. Continuous beams were verified for shear safety with appropriate stirrups , and columns were successfully evaluated and proven safe against both uni-axial and bi-axial bending moments using Bresler's empirical formulation.
- **Foundation:** A 1.0m x 1.0m rectangular shallow foundation was designed. It was validated against a safe soil bearing capacity of 200 KN/m² and successfully passed all critical checks for both one-way shear and punching (two-way) shear.

Ultimately, this manual design process proves that high-quality, structurally sound housing can be achieved economically. The resulting structural design guarantees minimal material wastage and provides a safe, permanent, and dignified home that fulfills the socio-economic goals of the Indiramma Illu initiative.

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APPENDIX

A = total area

A_c = area of concrete

A_g = gross area of the section

A_s= minimum area of tension reinforcement.

A_{sc} = area of steel in compression.

A_{st} = area of steel in tension

$A_{st\ max}$ = maximum area of steel required.

$A_{st\ min}$ = minimum area of steel required.

A_{sv} = total cross-sectional area of stirrup legs effective in shear.

B = width of the section

B_o = perimeter of critical section

b = width of the column

b_e = Equivalent breadth of the footing

DL = dead load

d = effective depth

d_o = D = overall depth

d' = Effective cover.

E = Young modules of elasticity

e = eccentricity

f_{ck} = Compressive Strength of concrete after 28 days – MPa or N/mm2

$F_{e\ 415}$ = Grade of steel

f_s = Steel stress of service load.

f_y = Yield strength of steel – MPa or N/mm2

H = Total height of wall.

K = stiffness of section.

L = length of the section

L_{eff} = Effective Length.

LL = live load

L_x = length of the section in X-direction.

L_y = length of the section in Y-direction

M = bending moment

M_{20} = Grade of concrete

M_u = moment of resistance of a section without compression reinforcement

$M_{u,limit}$ = Limiting moment of resistance

M_x = bending moment along X-axis

M_y = bending moment along Y-axis

P = axial load on column

P_u = factored axial load on column

q = net upward pressure

q_u = ultimate net upward pressure

S = spacing

SBC = The safe bearing capacity of soil

S_v = stirrup Spacing along the Length of the member.

T = Tension force.

τ_c = Design shear strength of concrete.

$\tau_{c\ max}$= maximum shear stress

τ_v = nominal shear stress

V = shear force

V_u =Factored shear force

V_{us} =Factored shear force of stirrup

W = total load

x_u = depth of neutral axis

$x_{u\ max}$ = limiting value of depth of neutral axis

Z = Cantilever Projection

$\emptyset$ = diameter of bar